

DATA MODELS COMPARISON

HIERARCHICAL MODEL

- Tree structure
- Parent-child relationship
- One-to-many
- Single parent only
- Example: File system
- Advantage: Simple and fast
- Disadvantage: Rigid structure

NETWORK MODEL

- Graph structure
- Many-to-many relationship
- Multiple parents allowed
- Complex connections
- Example: Transport network
- Advantage: Flexible
- Disadvantage: Complex to use

RELATIONAL MODEL

- Table structure
- Rows and columns
- Uses SQL
- Primary and foreign keys
- Example: MySQL, Oracle
- Advantage: Easy to use
- Disadvantage: Can be slow